From Lie Symmetry to Machine Learning: a conceptual map for semi-analytical PDE solving

LieML project — scoping note

July 5, 2026

Abstract

This note is a self-contained, beginner-friendly map of the ideas behind the project *"Semi-Analytical and Machine-Learning Methods for Solving PDEs Using Lie Symmetry Analysis."* It assumes little prior exposure to Lie groups or to how the symmetries of a partial differential equation (PDE) are classified. It builds the picture in three layers — continuous symmetry, symmetry *reduction* (your "variable compression" intuition, made precise), and the PDE–Lie–ML triangle — and then answers two questions directly: (i) *should you study finite groups of Lie type and their topology?* and (ii) *is a "combinatorial method" useful for analysing PDE–Lie, ML–Lie, and PDE–ML correlations?* Throughout, the abstract claims are anchored to small computations that have already been run and verified for this project (Burgers, the heat equation, the porous-medium equation, and Fisher–KPP).

Contents

1 The one-sentence picture **2**

2 Lie groups and Lie algebras, gently **2**
2.1 Continuous symmetry: one-parameter groups . . . 2
2.2 The infinitesimal generator: linearise the knob . . . 3
2.3 What a manifold is — and what makes a group a Lie group . . . 3
2.4 A concrete model to hold onto: matrix (linear) Lie groups . . . 6
2.5 A catalogue of the classical groups (and their special/unitary cases) . . . 6
2.6 A visual legend of the groups . . . 8

3 How a PDE knows its own symmetries **8**
3.1 Prolongation and the invariance condition . . . 8
3.2 Determining equations: a linear system the computer solves . . . 8

4 "Variable compression" made precise: symmetry reduction **8**
4.1 Finding the invariants: the characteristic system . . . 10
4.2 Why the whole algebra matters, not just one generator . . . 11

5 Classifying the symmetries: where combinatorics legitimately appears **11**

6 Question 1: finite groups of Lie type and topology — do you need them? **12**

7 Question 2: the combinatorial method and the PDE–Lie–ML triangle **12**
7.1 The unifying object . . . 13
7.2 Two genuine layers of structure . . . 13
7.3 What "correlations across the triangle" can concretely mean for you . . . 13

8 How this scopes your project **14**

9 A staged reading plan (what to learn, in order) **14**

10 The deepest finding: derivative and Lie symmetry are one object **15**
10.1 Generator versus flow: the primitive/group-like duality . . . 15
10.2 The atom $[d/dx, x] = 1$ and why the derivative is canonical . . . 15
10.3 Scaling is the Cartan grading; reduction is projection . . . 16
10.4 Prolongation is a representation; invariants are highest-weight vectors . . . 16
10.5 Fundamental PDEs realise classical algebras; ML equivariance is the same statement 16

11 Exchange diagrams (commutative diagrams) for the basic definitions **17**

12 Foundations: sheaves, schemes, jets, D-modules **19**
12.1 Presheaves and sheaves: local-to-global . . . 19
12.2 Affine schemes: "geometry is commutative algebra" . . . 19
12.3 Jet bundles and jet schemes: where derivatives live . . . 20
12.4 Differential operators and D-modules . . . 20
12.5 Hopf algebras and the primitive/group-like dictionary . . . 20
12.6 Differential invariants, moving frames, and diffieties . . . 20

Glossary **21**

References (advanced sections) **23**

1 The one-sentence picture

A PDE is hard for two reasons: it has *many independent variables* and an *enormous solution space*. A continuous symmetry is a knob you can turn that sends solutions to solutions. The central fact of the subject is:

Key idea. Each independent "symmetry knob" lets you *remove one independent variable* from the problem. Turning the knobs in the right order *compresses* a PDE in several variables down to an ordinary differential equation (ODE) in one variable — which you can often solve exactly or with a short series.

So your instinct is exactly right: **Lie theory is a "variable-compression" tool.** The rest of this note explains *why* that works, *how* the symmetries are found and classified, and where machine learning (ML) fits in.

A familiar special case you already know. Dimensional analysis (the Buckingham Π-theorem) is precisely symmetry reduction for the *scaling* symmetry: it replaces several dimensional variables by fewer dimensionless groups. Lie theory is the systematic generalisation to *all* continuous symmetries, not just scalings.

2 Lie groups and Lie algebras, gently

2.1 Continuous symmetry: one-parameter groups

Definition 2.1 (One-parameter group of transformations)**.** A family of maps g_ε depending smoothly on a real parameter ε with $g_0 = \mathrm{id}$ and $g_{\varepsilon_1} \circ g_{\varepsilon_2} = g_{\varepsilon_1+\varepsilon_2}$.

Examples on the plane (x, t) or on (x, t, u):

- **Translation:** $g_\varepsilon : (x) \mapsto (x + \varepsilon)$.
- **Scaling:** $g_\varepsilon : (x) \mapsto (e^\varepsilon x)$.
- **Rotation:** $g_\varepsilon : (x, t) \mapsto (x\cos\varepsilon - t\sin\varepsilon,\ x\sin\varepsilon + t\cos\varepsilon)$.

A **Lie group** is, informally, a group that is simultaneously a smooth surface (a manifold), so that "move a little" makes sense. The examples above are the building blocks $(\mathbb{R}, +)$, $(\mathbb{R}_{>0}, \times)$, and $SO(2)$.

2.2 The infinitesimal generator: linearise the knob

The decisive simplification is to look at a transformation *near* $\varepsilon = 0$. Differentiating the orbit gives a velocity field — a **vector field** — called the *infinitesimal generator*:

$$X \;=\; \xi(x,t,u)\,\frac{\partial}{\partial x} + \tau(x,t,u)\,\frac{\partial}{\partial t} + \phi(x,t,u)\,\frac{\partial}{\partial u}, \qquad (\xi,\tau,\phi) = \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} g_\varepsilon.$$

The set of all such generators, equipped with the commutator bracket $[X, Y] = XY - YX$, is the **Lie algebra** $\mathfrak{g}$ of the group.

Key idea (Why the algebra is the workhorse)*.* The group law is *nonlinear*, but the generators form a *linear* space. Lie's theorems say the whole local group is recovered from its generators by the exponential (the flow of the vector field),

$$g_\varepsilon = \exp(\varepsilon X).$$

So we trade a hard nonlinear search over groups for a *linear* computation over generators. This is exactly why symmetry analysis is *algorithmic*.

Remark 2.1 ("Finite-dimensional" means the *algebra*, not a finite group)*.* When people say a PDE has a "finite-dimensional symmetry algebra" they mean $\mathfrak{g}$ is a finite-dimensional *vector space* (e.g. dimension 5 for Burgers below). This has nothing to do with *finite groups*. We return to this important point in §6.

2.3 What a manifold is — and what makes a group a Lie group

We have been saying "a Lie group is a group that is also a smooth manifold." That one sentence carries the whole subject, so we now unpack it word by word, starting from nothing more than the idea of a continuous map. Everything in §2.1–2.2 — "move a little", the velocity field X, the exponential — is legitimate *only because* the underlying object is a smooth manifold, a space on which calculus makes sense. (For the topology in full, the standard gentle reference is Lee, *Introduction to Smooth Manifolds*; the `lee_smooth_manifolds` PDF in `ref/`.)

Step 1 — continuity and homeomorphism.

Definition 2.2 (Topological space; homeomorphism)**.** A **topological space** is a set together with a chosen collection of "open" subsets — a declaration of which points count as *near* each other — just enough structure to say when a map is **continuous** (preimages of open sets are open). A **homeomorphism** is a continuous bijection $f : X \to Y$ whose inverse f^{-1} is also continuous. When one exists, X and Y are **homeomorphic**: "the same up to continuous deformation," with no cutting and no gluing.

Example 2.1 (Topologically the same, or not)**.** The open interval $(-1, 1)$ is homeomorphic to the whole line $\mathbb{R}$ (e.g. via $x \mapsto \tan(\pi x/2)$); the proverbial coffee mug is homeomorphic to a doughnut. But a circle is *not* homeomorphic to a line segment — you would have to cut it. Homeomorphism is the right notion of "same shape" *before* any distance or smoothness is imposed.

Step 2 — locally Euclidean.

Definition 2.3 (Locally Euclidean)**.** A space M is **locally Euclidean of dimension** n if every point $p \in M$ has an open neighbourhood U that is homeomorphic to an open subset of $\mathbb{R}^n$. In words: *up close, M looks like flat $\mathbb{R}^n$.*

This is the Earth-looks-flat-locally idea made exact: the globe is curved, yet any city fits on a flat street map. For a connected space the number n is the same at every point; it is the **dimension** of M.

Step 3 — charts and atlases.

Definition 2.4 (Chart, atlas, transition map)**.** A **chart** is a homeomorphism $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^n$ from an open set $U \subseteq M$; it is a local *coordinate system*, attaching numbers $(x^1, \ldots, x^n)$ to the points of U. An **atlas** is a family of charts whose domains cover M. Where two charts overlap, the **transition map** $\varphi_\beta \circ \varphi_\alpha^{-1}$ re-expresses one set of coordinates in terms of the other (Fig. 1).

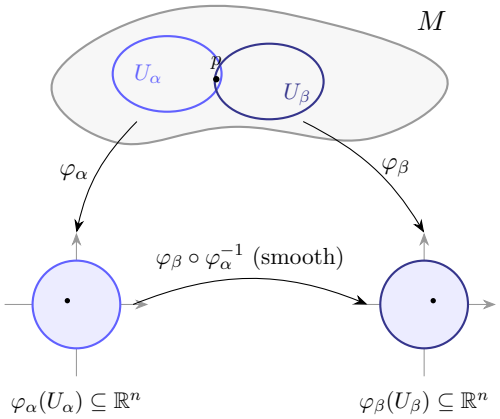


Figure 1: **A manifold and its charts.** Each point of the curved space M has a neighbourhood $(U_\alpha,\ U_\beta)$ carried by a chart $(\varphi_\alpha,\ \varphi_\beta)$ onto a flat patch of $\mathbb{R}^n$ — this is what *locally Euclidean* means. On the overlap (containing p) the two coordinate systems are related by the *transition map* $\varphi_\beta \circ \varphi_\alpha^{-1}$; requiring every such transition to be infinitely differentiable is exactly what makes M a *smooth* manifold.

Step 4 — topological and smooth manifolds.

Definition 2.5 (Manifold)**.** A **topological manifold** of dimension n is a space that is (i) *locally Euclidean* of dimension n, (ii) *Hausdorff* (any two points lie in disjoint open sets — no "doubled" points), and (iii) *second-countable* (its topology has a countable base — it is not pathologically large). It is a **smooth (C^∞) manifold** if it carries an atlas whose transition maps are all infinitely differentiable.

Key idea (Why "smooth" is the whole point)*.* Smooth transition maps are exactly what let calculus pass from one chart to another *unambiguously*: whether a function is differentiable, and what its derivative is, cannot depend on which coordinate system you happened to pick. On a smooth manifold one can therefore define smooth functions, differentiate them, and — decisively for us — define *tangent vectors* and *vector fields* without ever leaving the abstract space. The generator X of §2.2 lives here.

Definition 2.6 (Tangent space)**.** The **tangent space** T_pM at a point p is the n-dimensional vector space of velocities $\dot\gamma(0)$ of smooth curves γ passing through p. A **vector field** chooses one tangent vector $X_p \in T_pM$ at every point, smoothly — precisely the $\xi\,\partial_x + \tau\,\partial_t + \phi\,\partial_u$ of §2.2.

Step 5 — a Lie group.

Definition 2.7 (Lie group)**.** A **Lie group** is a set G that is *simultaneously*

- a **group** — an associative product with an identity e and inverses — and
- a **smooth manifold**,

with the two structures *compatible*: multiplication $G \times G \to G,\ (g,h) \mapsto gh$ and inversion $G \to G,\ g \mapsto g^{-1}$ are *smooth* maps. Its **dimension** is its dimension as a manifold — the number of independent continuous parameters (the "knobs" of §2.1).

Key idea (Where the Lie algebra comes from)*.* Because G is a smooth manifold, it has a tangent space at the identity, and *that* is the Lie algebra: $\mathfrak{g} := T_eG$. The infinitesimal generators of §2.2 are exactly the elements of T_eG, and the bracket $[X,Y]$ measures how the group multiplication fails to commute near e. Smoothness of the group operations is what makes the exponential $\exp : \mathfrak{g} \to G$ a local diffeomorphism — the rigorous form of "recover the local group from its generators" (§2.2). In particular $\dim G = \dim \mathfrak{g}$: the circle is 1-dimensional, $SO(3)$ is 3-dimensional, $GL(n,\mathbb{R})$ is n^2-dimensional.

Example 2.2 (The circle: the cleanest Lie group)**.** The unit circle $SO(2) = \{e^{i\theta}\}$ is a 1-dimensional smooth manifold (one angle coordinate), and angle addition $e^{i\theta}e^{i\phi} = e^{i(\theta+\phi)}$ is smooth — so it is a Lie group. It already illustrates why charts are unavoidable: *no single* chart covers the whole circle (any angle coordinate must jump somewhere), so the smallest atlas needs two overlapping arcs. Locally the circle is just $\mathbb{R}$; globally it closes up — it is *compact*.

Remark 2.2 (Manifold $\neq$ Lie group)*.* Every Lie group is a manifold, but most manifolds are *not* Lie groups. The 2-sphere S^2 carries no continuous group law at all (relatedly, you cannot comb a smooth nonvanishing vector field on it — the "hairy ball" theorem); among spheres only $S^1 = U(1)$ and $S^3 = SU(2)$ are Lie groups. The group structure is a genuine extra constraint on top of being a smooth space.

Step 6 — back to matrices. This abstract picture is realised concretely by the **matrix Lie groups** of the next subsection, which are the hands-on manifolds to keep in mind:

- $GL(n,\mathbb{R})$ is an *open* subset of the n^2-dimensional space of all real matrices (the condition $\det \neq 0$ is open), hence a manifold of dimension n^2 described by a single global chart.
- $O(n)$, $SO(n)$, $U(n)$, … are *level sets* cut out by smooth equations such as $A^\top A = I$: curved "surfaces" sitting inside matrix space, i.e. embedded submanifolds.

Key idea (Closed-subgroup theorem: why "closed" buys smoothness). Cartan's **closed-subgroup theorem** states that *any* subgroup of $GL(n,\mathbb{C})$ that is topologically closed is automatically a smooth embedded submanifold — hence a Lie group. This is exactly why the matrix definition of §2.4 needs only the single word *closed*: smoothness, the tangent space, and the exponential all come for free. Matrix Lie groups are therefore the most concrete manifolds to picture — surfaces in matrix space on which exp, the bracket, and representations become ordinary linear algebra.

2.4 A concrete model to hold onto: matrix (linear) Lie groups

The manifold definition above is correct but abstract. The fastest rigorous on-ramp — and the viewpoint of B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction* — is to make everything *matrices*.

Definition 2.8 (Matrix (linear) Lie group)**.** A **matrix Lie group** is a closed subgroup of the general linear group $GL(n,\mathbb{C})$ (invertible $n \times n$ complex matrices). Examples: $SL(n)$ (determinant 1), $O(n)$ and $SO(n)$ (orthogonal / rotations), $U(n)$ and $SU(n)$ (unitary), and the Heisenberg and Euclidean groups.

In this model the three abstract notions of §2.1–2.2 become elementary linear algebra:

- **Group → algebra via the matrix exponential.** $\exp(A) = \sum_{k\geq 0} A^k/k!$, and a one-parameter subgroup is $g(t) = \exp(tX)$ — the matrix analogue of $g_\varepsilon = \exp(\varepsilon X)$.
- **The Lie algebra is a space of matrices.** $\mathfrak{g} = \{\, X : \exp(tX) \in G \text{ for all } t \in \mathbb{R} \,\}$, a real vector space closed under the *commutator* $[X,Y] = XY - YX$. Concretely $\mathfrak{so}(n)$ = skew-symmetric matrices, $\mathfrak{sl}(n)$ = traceless matrices, $\mathfrak{su}(n)$ = skew-Hermitian *and* traceless matrices (skew-Hermitian alone is $\mathfrak{u}(n)$). Note $O(n)$ and $SO(n)$ share the *same* algebra $\mathfrak{so}(n)$ — the algebra only sees the identity component.
- **The bracket measures non-commutativity (Baker–Campbell–Hausdorff).** $\exp(X)\exp(Y) = \exp\big(X + Y + \frac{1}{2}[X,Y] + \cdots\big)$, so the single bracket already controls how the group fails to commute — the precise sense in which "the algebra knows the local group."

Key idea (Representations — the bridge to ML). A **representation** is a homomorphism $\rho : G \to GL(V)$ (or $\mathfrak{g} \to \mathfrak{gl}(V)$): it records *how the group acts linearly* on a vector space V. PDE solution spaces and ML feature spaces are exactly such V's. Then "symmetry of a PDE" and "equivariance of a network" become the *same* statement — a map that intertwines two representations, $\rho_{\text{out}}(g)\,\Phi = \Phi \circ \rho_{\text{in}}(g)$. Representation theory (the third pillar of Hall's book) is the common language of the PDE–Lie–ML triangle (§7).

Remark 2.3 (How the matrix picture connects to the vector-field picture). For PDE symmetry the group acts on (x,t,u)-space and the generators are *vector fields* (§2.2), whose action can be *nonlinear*. Matrix Lie groups are the cleanest *linear* special case — the right place to first meet exp, the bracket, BCH and representations — and prolongation (letting X act on derivatives) is itself representation-theoretic: the group acts linearly on the "jet" of derivatives. Linear actions also dominate the ML side (equivariant layers), so Hall's matrix route is the most efficient bridge for this project.

2.5 A catalogue of the classical groups (and their special/unitary cases)

The matrix groups of §2.4 fall into a few families, each defined by the *structure it preserves*: a volume, a real inner product, a Hermitian form, or a symplectic form. Two prefixes recur. **Special** (S) means *determinant* 1 (volume/orientation preserving); **unitary** (U) means *preserves a Hermitian form* (the complex analogue of orthogonal). Every symmetry algebra met in PDE analysis is a subalgebra of one of these, or of an *inhomogeneous* (translation) extension of one (Table B). Each group below is also drawn as an icon in Fig. 2.

Table A. The classical (linear) Lie groups.

Group	Preserves / defining condition	$\dim_{\mathbb{R}}$	Compact
$GL(n,\mathbb{R})$	invertible real matrices ($\det \neq 0$)	n^2	no
$SL(n,\mathbb{R})$	+ volume: $\det = 1$	n^2-1	no
$GL(n,\mathbb{C})$	invertible complex matrices	$2n^2$	no
$SL(n,\mathbb{C})$	+ $\det = 1$	$2(n^2-1)$	no
$O(n)$	real symmetric form: $A^{\top}A = I$ (lengths)	$\frac{1}{2}n(n-1)$	yes
$SO(n)$	+ $\det = 1$ (rotations)	$\frac{1}{2}n(n-1)$	yes
$O(p,q)$	real form of signature (p,q)	$\frac{1}{2}(p{+}q)(p{+}q{-}1)$	no
$SO^{+}(1,3)$	**Lorentz**, proper orthochronous	6	no
$O(n,\mathbb{C})$, $SO(n,\mathbb{C})$	complex orthogonal	$n(n-1)$	no
$U(n)$	Hermitian form: $A^{*}A = I$	n^2	yes
$SU(n)$	+ $\det = 1$	n^2-1	yes
$U(p,q)$	Hermitian form of signature (p,q)	$(p{+}q)^2$	no
$SU(p,q)$	+ $\det = 1$	$(p{+}q)^2-1$	no
$Sp(2n,\mathbb{R})$	real symplectic (antisymmetric) form	$n(2n{+}1)$	no
$Sp(n) = USp(2n)$	compact symplectic $= Sp(2n,\mathbb{C}) \cap U(2n)$	$n(2n{+}1)$	yes
$Sp(2n,\mathbb{C})$	complex symplectic	$2n(2n{+}1)$	no

Coincidences: $SO(2) = U(1)$ (the circle); $Sp(2,\mathbb{R}) = SL(2,\mathbb{R})$; $Sp(1) = SU(2)$; double covers $SU(2) \to SO(3)$ and $SL(2,\mathbb{C}) \to SO^{+}(1,3)$ link the unitary world to rotations and Lorentz boosts. $O(n)$ and $SO(n)$ share the algebra $\mathfrak{so}(n)$ (the algebra sees only the identity component).

Table B. Inhomogeneous, affine, and PDE-symmetry groups ($\rtimes$ = semidirect product).

Group	Structure	$\dim_{\mathbb{R}}$	Role in PDE / Lie analysis
$T(n)$	$(\mathbb{R}^n,+)$, translations (abelian)	n	generators ∂_i; autonomous PDEs
$(\mathbb{R}_{>0},\times)$	dilation / scaling	1	similarity reduction; Buckingham Π
$E(n)$	$\mathbb{R}^n \rtimes O(n)$, **Euclidean**	$\frac{1}{2}n(n{+}1)$	isotropic homogeneous PDEs (Laplace)
$SE(n)$	$\mathbb{R}^n \rtimes SO(n)$, proper motions	$\frac{1}{2}n(n{+}1)$	orientation-preserving case
Gal(d)	**Galilei**: tr., rot., boosts	$\frac{1}{2}d(d{-}1){+}2d{+}1$	heat/diffusion; non-relativistic
Sch(d)	**Schrödinger** $= SL(2,\mathbb{R}) \ltimes$ Gal	6 ($d{=}1$)	max. symmetry of free heat (§10)
ISO$(1,3)$	$\mathbb{R}^{1,3} \rtimes O(1,3)$, **Poincaré**	10	wave, Klein–Gordon, Maxwell, Dirac
Conf(p,q)	$\cong O(p{+}1,q{+}1)$, **conformal**	$\frac{1}{2}(p{+}q{+}2)(p{+}q{+}1)$	Laplace/wave (§12)
Aff(n)	$\mathbb{R}^n \rtimes GL(n,\mathbb{R})$, affine	$n(n{+}1)$	linear changes of variable
H_n	Heisenberg, $[P_i,Q_j]{=}\delta_{ij}Z$	$2n{+}1$	the atom $[\partial,x]{=}1$ (§10)
$SL(2,\mathbb{R})$	$\sim PSL(2,\mathbb{R})$, Möbius	3	the $\mathfrak{sl}_2$ of §10; 2-D conformal

Here d is the number of *space* dimensions. The geometric Galilei group has $\dim \mathrm{Gal}(d) = \frac{1}{2}d(d-1) + 2d + 1$ ($= 10$ for d=3); the mass central extension (the *Bargmann* group) adds 1. The centrally-extended Schrödinger group $\mathrm{Sch}(d) = SL(2,\mathbb{R}) \ltimes$ Bargmann has dim $= \frac{1}{2}d(d-1) + 2d + 4$, i.e. 6 for d=1 (the $\mathfrak{sl}_2\ltimes$ Heisenberg of §10) and 13 for d=3. The maximal point-symmetry group of a fundamental PDE is one of these: heat → Schrödinger, wave/Laplace → conformal, Burgers → the projective group of Example 3.1.

Remark 2.4 (Two facts worth carrying). (i) In *two* dimensions the conformal group is *infinite*-dimensional (the Witt/Virasoro algebra) — the structural reason 2-D problems are so often integrable; in ≥ 3 dimensions it is the finite group $O(p{+}1, q{+}1)$ above. (ii) Compactness is decisive for machine learning: the PDE groups that matter (translations, scalings, boosts, the Lorentz/Galilei/conformal groups) are *non-compact*, which is exactly what makes equivariant architectures hard and motivates the canonicalisation approaches of §7.

2.6 A visual legend of the groups

Each icon below shows what the group *moves*: the set it acts on and the invariant it preserves. Grey marks the scaffolding (axes, asymptotes, mirrors, or the *original* position of a reference object); blue marks the group's *action* or the preserved object. Dashed strokes are light-cone asymptotes, reflection mirrors, or before-images. Complex ($\mathbb{C}$) and pseudo cases (signature (p, q)) are the higher-dimensional analogues of the real prototypes drawn here.

3 How a PDE knows its own symmetries

3.1 Prolongation and the invariance condition

A symmetry of a PDE $\Delta = 0$ is a transformation that maps solutions to solutions. Because Δ involves derivatives $u_x, u_t, u_{xx}, \ldots$, we must also know how the generator X acts on derivatives. That extension is the **prolongation** $\mathrm{pr}^{(k)}X$, computed by a fixed (if tedious) formula. The test for a symmetry is then beautifully simple:

$$\boxed{\mathrm{pr}^{(k)}X(\Delta)\big|_{\Delta=0} = 0.} \tag{1}$$

Read (1) as: "moving along the symmetry does not push you off the solution set."

3.2 Determining equations: a linear system the computer solves

Plugging an *unknown* generator $X = \xi\partial_x + \tau\partial_t + \phi\partial_u$ into (1) and demanding it hold for *all* solutions forces the coefficients of the independent derivative terms to vanish separately. The result is the **determining equations**: a large but *linear*, *overdetermined* system of PDEs for (ξ, τ, ϕ). Its solution space *is* the symmetry algebra $\mathfrak{g}$. This is mechanisable (Maple's `PDEtools`, `GeM`, `SADE`; Mathematica's `SYM`; and, for this project, `sympy`).

Example 3.1 (Verified for this project)**.** For viscous Burgers $u_t + uu_x = \nu u_{xx}$, our `sympy` prolongation engine confirms a **5-dimensional** algebra (invariance residual $\equiv 0$ for each generator):

$$\partial_t, \quad \partial_x, \quad \underbrace{t\,\partial_x + \partial_u}_{\text{Galilean}}, \quad \underbrace{2t\,\partial_t + x\,\partial_x - u\,\partial_u}_{\text{scaling}}, \quad \underbrace{t^2\partial_t + tx\,\partial_x + (x - tu)\partial_u}_{\text{projective}}.$$

(File: `symm/feasibility_demo.py`.) For the linear heat equation $u_t = u_{xx}$ the point algebra is **6-dimensional** *plus* an infinite-dimensional piece $\beta(x,t)\partial_u$ coming from linear superposition (any solution can be added). Burgers is the heat equation in disguise (the Cole–Hopf transform), which is why their symmetry structures are cousins.

Classical (linear) groups — Table A



$GL(n,\mathbb{R})$
$\det \neq 0$

$SL(n,\mathbb{R})$
area fixed

$GL(n,\mathbb{C})$
$\mathbb{C}$-*scale*

$SL(n,\mathbb{C})$
$\det = 1$



$O(n)$
$\|x\|$; + *flip*

$SO(n)$
$\|x\|$, $\det = 1$

$O(p,q)$
$x^2 - y^2$

$SO^+(1,3)$
Lorentz



$O, SO(n,\mathbb{C})$
complex $\perp$

$U(n)$
$|z|^2$

$SU(n)$
$|z|^2$, $\det = 1$

$U(p,q)$
$|z_1|^2 - |z_2|^2$



$SU(p,q)$
$+ \det = 1$

$Sp(2n,\mathbb{R})$
$dq \wedge dp$

$Sp(n)$
compact, $USp(2n)$

$Sp(2n,\mathbb{C})$
$\mathbb{C}$-*symplectic*

Inhomogeneous & PDE-symmetry groups — Table B



$T(n)$
$x \mapsto x + a$

$(\mathbb{R}_{>0}, \times)$
$x \mapsto \lambda x$

$E(n)$
$Rx+a$ (+*flip*)

$SE(n)$
$\det R = 1$



Gal(d)
$(x+vt,\ t)$

Sch(d)
$(\lambda x,\ \lambda^2 t)$

ISO$(1,3)$
Δs^2, *Poincaré*

Conf(p,q)
angles



Aff(n)
$Ax + b$

H_n
$[x,p] = 1$

$SL(2,\mathbb{R})$
Möbius on $\mathbb{H}$

Figure 2: **Visual legend of the catalogue** (§2.5). Each icon shows the set the group acts on (grey: axes / asymptotes / mirror / before-image; blue: the action or preserved object) and the invariant it fixes. The complex ($\mathbb{C}$) and pseudo-signature (p,q) entries are the higher-dimensional analogues of these real 2-D prototypes; the combined $O, SO(n,\mathbb{C})$ panel covers both the full and special complex-orthogonal groups.

4 "Variable compression" made precise: symmetry reduction

This is the heart of your intuition. Fix one generator X. An **invariant** of X is a function I with $X(I) = 0$. For a one-parameter group acting on (x, t, u) there are exactly two functionally independent invariants: a **similarity variable** z (often a function of the independent variables x, t) and a dependent invariant. Looking for solutions that are themselves invariant under X forces the ansatz

$$u(x,t) = \big(\text{prefactor}\big) \cdot U\big(z(x,t)\big),$$

and substituting it into the PDE collapses two independent variables (x, t) into one (z): a PDE becomes an **ODE**.

4.1 Finding the invariants: the characteristic system

The paragraph above asserts that two invariants exist; here is the mechanism that produces them — all a computer (or a hand calculation) actually needs. The generator $X = \xi\partial_x + \tau\partial_t + \phi\partial_u$ is the velocity field of a flow on (x, t, u)-space, and an invariant is anything constant along that flow. Constancy is exactly the equality of the following ratios, the **characteristic system**:

$$\boxed{\frac{dx}{\xi} = \frac{dt}{\tau} = \frac{du}{\phi}} \tag{2}$$

Every independent first integral of (2) is one invariant; on the three coordinates (x, t, u) there are two — the similarity variable z, and one that fixes the profile.

Example 4.1 (Fisher–KPP, worked in full and re-verified)**.** Combine Fisher's two basis symmetries into $X = c\,\partial_x + \partial_t$ (so $\xi = c,\ \tau = 1,\ \phi = 0$). The characteristic system (2) becomes $\dfrac{dx}{c} = \dfrac{dt}{1} = \dfrac{du}{0}$, and its two invariants can be read straight off:

- $\dfrac{du}{0}$ forces $du = 0$: the value u rides along unchanged — the profile U;
- $\dfrac{dx}{c} = \dfrac{dt}{1}$ integrates to $x - ct = \text{const}$ — the similarity variable $z = x - ct$.

An X-invariant solution therefore has the form $u(x, t) = U(z)$ with $z = x - ct$, and every derivative becomes a z-derivative times a fixed factor,

$$u_t = -c\,U', \qquad u_x = U', \qquad u_{xx} = U''.$$

Fisher's equation $u_t = u_{xx} + u(1 - u)$ then collapses to

$$-c\,U' = U'' + U(1 - U) \qquad \Longleftrightarrow \qquad \boxed{U'' + c\,U' + U(1 - U) = 0\,,}$$

an ODE in z alone: the two independent variables have become one. (Re-verified for this note — substituting $u = U(x - ct)$ leaves an expression whose only free symbols are c and z, and the engine independently certifies ∂_x, ∂_t as symmetries with residual $\equiv 0$; files `symm/sym_engine/reduce.py`, `symm/fisher_kpp_reduction.py`.)

Key idea (The compression, quantified). A PDE on $\mathbb{R}^2$ (variables x, t) reduces to an ODE on $\mathbb{R}$ (variable z). More generally, an r-parameter solvable subgroup reduces the number of independent variables by r. The infinite solution space collapses onto the finite-parameter family of *group-invariant* (similarity) solutions.

Example 4.2 (Three reductions we verified)**.**

- **Burgers, travelling wave** (combine ∂_t, ∂_x): $z = x - ct$, $u = U(z)$ gives $\nu U'' = (U - c)U'$, solved exactly by the shock $U = c - k\tanh\left(\frac{kz}{2\nu}\right)$ (residual 0).
- **Porous medium** $u_t = (u^m u_x)_x$ (scaling): $u = t^{-\alpha}F(\eta)$, $\eta = xt^{-\beta}$, (mass conservation $\Rightarrow \alpha = \beta$, scaling $\Rightarrow \beta = \frac{1}{m+2}$) gives the Barenblatt ODE $(F^m F')' + \beta(\eta F)' = 0$; the closed-form self-similar profile checks out for $m = 1, 2, 3$. (This is the textbook example of "compression by scaling".)
- **Fisher–KPP** $u_t = u_{xx} + u(1 - u)$ (travelling wave): $z = x - ct$ gives the ODE $U'' + cU' + U(1 - U) = 0$ (derived in full in Example 4.1), *non-integrable for generic* c — with a closed-form Ablowitz–Zeppetella front only at $c = 5/\sqrt{6}$. (Files: `symm/feasibility_demo.py`, `symm/fisher_kpp_reduction.py`.)

4.2 Why the whole algebra matters, not just one generator

One generator gives one reduction. The *algebra* — the full solution space of the determining equations — adds two things more, and they are exactly the two quantities the engine computes and that we checked against the literature.

Its dimension is the size of the menu. Each inequivalent one-dimensional subalgebra yields a different similarity variable, hence a different reduction. Fisher's algebra is two-dimensional and essentially offers the single travelling wave above; Burgers' five-dimensional algebra (Example 3.1) offers many reductions, and its very size reflects that Burgers is linearisable (Cole–Hopf); the porous medium's four-dimensional algebra contains genuine *scalings*, which is why it admits self-similar (Barenblatt) solutions. Bigger algebra, richer menu.

Its bracket structure decides how far you can go. After a reduction by X, the other symmetries do not vanish: those compatible with X (its normaliser) descend to symmetries of the reduced *ODE*. If $\mathfrak{g}$ is **solvable**, Lie's theorem then integrates that ODE by successive quadratures — one symmetry peeled off per order until an explicit answer appears. Fisher's reduced ODE $U'' + cU' + U(1 - U) = 0$ still carries a z-translation symmetry, and that leftover is exactly what lets one study the fronts in the (U, U') **phase plane**. (Testing solvability — the derived and lower-central series of $\mathfrak{g}$ — is the job of `symm/sym_engine/algebra.py`.)

Remark 4.1 (Two branches, and the honest infinite case). The engine's `reduce.py` implements the two commonest generator types directly: a *translation* branch (travelling waves, $z = x - ct$) and a *scaling* branch (self-similar solutions, $z = x/t^a$, $u = t^b U$); which one fires is set by the generator supplied. When the algebra is *infinite*-dimensional — the heat and transport equations, which the validation flags as `stalled` rather than pinning a finite dimension — its generators include the superposition symmetry $u \mapsto u + \beta$ shared by every linear PDE. That is compression of a different kind (linearity restated as symmetry), and the engine reports it as infinite instead of forcing a finite ODE.

When the reduced ODE is not solvable in closed form (Fisher–KPP, Blasius), we close it with a *semi-analytical series* (homotopy analysis) and, if needed, a small neural-network correction — which is where ML enters. The project's contribution is to attach a *convergence certificate* to that series; see the research plan in `plan.txt`.

5 Classifying the symmetries: where combinatorics legitimately appears

Once you have the algebra $\mathfrak{g}$ (say $\dim \mathfrak{g} = 5$ for Burgers) you can reduce by *any* subgroup — but infinitely many choices give *equivalent* reductions. The right bookkeeping is:

Definition 5.1 (Optimal system)**.** A list of one-dimensional subalgebras of $\mathfrak{g}$, one representative per conjugacy class under the group's *adjoint action*. Each class yields one genuinely inequivalent reduction; the rest are duplicates.

Building the optimal system is a finite, *combinatorial/algebraic* classification problem (commutator table $\to$ adjoint action $\to$ orbits of subalgebras). This is the first place a "combinatorial method" is not only useful but standard. Two further classification ideas:

- **Group classification:** within a family of equations (e.g. $u_t = (K(u)u_x)_x$ for various K), classify which K admit extra symmetry. A classification over a parameter space.
- **Structure of $\mathfrak{g}$ itself:** solvable / nilpotent / semisimple; semisimple algebras are classified by *root systems* (the Cartan classification). For PDE work the algebras are usually small, so this deep machinery is rarely needed — good to recognise, not to master first.

6 Question 1: finite groups of Lie type and topology — do you need them?

This is a terminology trap worth clearing up carefully, because it determines what you should and should not spend time on.

Object	What it is / where it lives
Lie group (continuous)	A group that is also a smooth manifold; real or complex. **This is what PDE symmetry uses**: translations, scalings, Galilean boosts, rotations, etc.
Finite-dimensional Lie algebra	The linear space of generators (e.g. dim = 5 for Burgers). "Finite" here means finite-*dimensional*, **not** a finite group. The object you compute with daily.
Finite groups of Lie type	Finite groups such as $PSL_n(q)$, built from algebraic groups over a *finite field* $\mathbb{F}_q$ (Chevalley groups). Central to the Classification of Finite Simple Groups — a different subject. **Essentially unused in PDE symmetry analysis.**

Verdict. **Do not detour into finite groups of Lie type for this project.** They belong to finite/algebraic group theory and will not help you reduce or solve PDEs. Invest instead in (a) continuous (local) Lie groups, (b) their finite-*dimensional* algebras, and (c) just enough manifold topology.

How much topology? Only a working amount:

- what makes a Lie group a smooth manifold, and the role of the *identity component* (symmetry analysis is inherently *local* — it lives near the identity, via the algebra);
- **compact vs. non-compact**, because the PDE symmetries you care about (scalings, translations, boosts) are typically *non-compact*, and that non-compactness is precisely what makes some equivariant-ML architectures struggle (see §7).

You do *not* need deep algebraic topology to start.

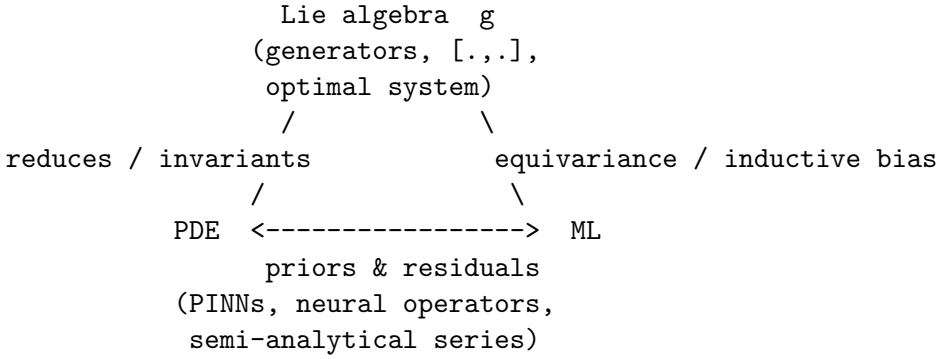
Where finite/compact groups *do* matter — on the ML side. Group-equivariant neural networks are built for translation (ordinary CNNs), finite rotation/reflection groups (group-equivariant CNNs), and compact groups (e.g. G-FNO). Handling *non-compact* groups (the PDE case) needs extra machinery (e.g. Lie-algebra canonicalisation, cf. `arXiv:2410.02698` in `ref/`). So a *little* compact/finite group representation theory pays off for the ML–Lie link — but the PDE–Lie reductions are driven by non-compact continuous groups.

7 Question 2: the combinatorial method and the PDE–Lie–ML triangle

Your idea — analyse PDE–Lie, ML–Lie, and PDE–ML *correlations* and look for shared threads or a "global symmetry" — is sound, provided we separate the parts that are concrete and high-value from the parts that are deep but easy to over-invest in.

7.1 The unifying object

Key idea. The Lie algebra $\mathfrak{g}$ is the *single object shared by all three corners*: it acts on the PDE's solution space, it organises the catalogue of reductions, and it is exactly the "inductive bias" an ML model can be told to respect. The "thread" you are sensing is real — it is $\mathfrak{g}$ itself.



7.2 Two genuine layers of structure

(A) The combinatorial / algebraic layer — concrete and recommended.

- **Subalgebra lattice and optimal systems** (§5): the finite catalogue of inequivalent reductions. Successive reductions form *chains* of subgroups — a poset. This is literally a combinatorial enumeration of "reduction tasks".
- **Differential elimination / Gröbner-type methods** (Rosenfeld–Gröbner; differential algebra) for *solving* the determining equations systematically. Combinatorial algebra doing real work.
- **Commutator tables and structure constants**: the finite data that drives everything above.

(B) The higher algebraic-geometric layer — powerful but optional.

- **Jet bundles**: the natural geometric space where PDEs, derivatives, and prolongations live. The honest "home" of (1).
- **Differential invariants and moving frames** (Fels–Olver): an invariant calculus that can *build* equations and ML features guaranteed to respect $\mathfrak{g}$.

- **Exterior differential systems / Cartan–Kähler**, and **D-modules** for linear PDEs: deep, but a research-career sink if pursued prematurely.

Verdict. For a limited-compute, results-oriented thesis: **yes, the combinatorial method is useful** — via layer (A), especially optimal systems (the combinatorics of *which* reductions to run) and differential elimination (*how* to solve the determining system). Treat layer (B) as enrichment: learn the language of jet bundles and differential invariants (they make the ML–Lie "feature" link rigorous), but do not start a thesis inside D-modules or exterior differential systems.

7.3 What "correlations across the triangle" can concretely mean for you

Two of the project's own contributions are exactly "cross-corner" findings, which shows the framing is productive:

- **The dead-generator phenomenon (PDE–Lie–ML).** Some generators that are perfectly good *reductions* contribute *zero ML training signal* (their symmetry-loss term collapses to 0 or a multiple of Δ). The project turns this into a design principle: a generator gives no ML signal exactly when its orbit is already "quotiented out" by working in invariant coordinates — so the failure mode disappears by construction (see `plan.txt`, §2/S4).
- **The optimal system as a combinatorial index of the whole pipeline.** Each conjugacy class in the optimal system ⇒ one reduced ODE ⇒ one semi-analytical series + certificate + ML corrector. The project is, structurally, a combinatorial enumeration over the subalgebra classes.

A caution on the speculative end. Treating "ML–Lie–PDE correlations" as a brand-new *global* algebraic-geometric invariant is a research hypothesis, not established theory. It is fine to hold as a long-range aim, but the *publishable, fundable* core is layer (A) plus the certified reduce–close–certify–correct pipeline.

8 How this scopes your project

Putting the layers in order of what to do:

Stage	Lie-theoretic content	Project artefact
Discover	solve determining equations (§3)	auto-find `g` (`sympy`)
Classify	optimal system (§5, layer A)	enumerate inequivalent reductions
Reduce	invariants → similarity ODE (§4)	verified ODEs (Burgers, porous, Fisher)
Close	homotopy-analysis series	interpretable series $U_M(z;\hbar)$
Certify	convergence ratio + error bound	the headline novelty
Correct	invariant-coordinate small PINN	ML remainder, CPU-scale

9 A staged reading plan (what to learn, in order)

1. **Symmetry methods for differential equations** (the practical core): Hydon, *Symmetry Methods for Differential Equations* (gentlest); then Olver, *Applications of Lie Groups to Differential Equations* (the standard reference: prolongation, determining equations, reduction, differential invariants); Bluman–Cheviakov–Anco for nonclassical/potential symmetries and conservation laws.

2. **Just-enough Lie groups/algebras:** the algebra–group correspondence, exponential map, commutators, solvability. **Start with Hall**, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction* — the gentlest *rigorous* route, because it works through concrete *matrix* Lie groups (matrix exponential, BCH, representations) before any heavy manifold theory (§2.4). In `ref/` you also have `Lie_Groups_Lectures` and `liegroups`; `lee_smooth_manifolds` is the manifold-topology backstop (use as reference, not cover-to-cover).
3. **Semi-analytical methods:** Liao on the Homotopy Analysis Method; the convergence-control parameter and the (conditional) convergence theorem.
4. **Symmetry + ML:** the two local papers — "Lie Point Symmetry and Physics-Informed Networks" and "Generalized Lie Symmetries in Physics-Informed Neural Operators" — plus the Fourier Neural Operator and Lie-algebra-canonicalisation papers in `ref/`.
5. **Optional enrichment:** jet bundles and moving frames (Olver), then differential algebra / Rosenfeld–Gröbner for determining systems.

10 The deepest finding: derivative and Lie symmetry are one object

Everything in §2–§6 quietly used one fact twice. We now state it outright, because it is the single most useful organising idea in the whole project: *the ordinary derivative and a Lie symmetry are not two subjects but one.* The container that holds both is the notion of a **derivation**.

Definition 10.1 (Derivation)**.** A **derivation** of the algebra $C^\infty(M)$ of smooth functions is a linear map $D : C^\infty(M) \to C^\infty(M)$ satisfying the *Leibniz rule*

$$D(fg) = D(f)\,g + f\,D(g).$$

Key idea (One containment, not two subjects)*.* The plain derivative $\dfrac{d}{dx}$ is exactly the Lie derivative $\mathcal{L}_{\partial_x}$ along the *constant* vector field ∂_x. A Lie symmetry generator is $\mathcal{L}_X$ for a *general* vector field X. Both obey the same Leibniz rule, so both are derivations. The derivative is the simplest symmetry generator; a symmetry is a derivative "bent" along a curved flow. This is why prolongation, invariance, and reduction can be *computed*: they are linear algebra over the space of derivations.

10.1 Generator versus flow: the primitive/group-like duality

Lie's exponential (§2.2) has an algebraic face. Inside the universal enveloping algebra of $\mathfrak{g}$ there is a coproduct Δ (a Hopf-algebra structure). A derivation is a **primitive** element, and its exponential is a **group-like** element:

$$\Delta(D) = D \otimes 1 + 1 \otimes D \qquad \xrightarrow{\exp} \qquad \Delta\big(e^{tD}\big) = e^{tD} \otimes e^{tD}.$$

The left equation *is* the Leibniz rule in disguise; the right equation *is* the statement "e^{tD} is an automorphism (a flow that respects products)." Passing primitive $\leftrightarrow$ group-like by exp / log is precisely Lie's generator $\leftrightarrow$ group correspondence (§2.4).

Example 10.1 (The derivative is the translation generator (Stone/Taylor))**.** For the constant field ∂_x, the group-like exponential acts by translation:

$$e^{a\,d/dx} f(x) = \sum_{n\geq 0} \frac{a^n}{n!}\, f^{(n)}(x) = f(x+a).$$

So "take a derivative" and "shift by a" are the infinitesimal and finite versions of one symmetry. This is the cleanest instance of the whole §10 thesis.

10.2 The atom $[d/dx,\, x] = 1$ and why the derivative is canonical

Multiplication by x and differentiation do not commute:

$$\left[\tfrac{d}{dx},\, x\right] = \tfrac{d}{dx} \circ x - x \circ \tfrac{d}{dx} = 1 \qquad \text{(the Heisenberg/Weyl relation } A_1\text{)}.$$

Adjoining the scaling operator $x\,d/dx$ closes this into a copy of $\mathfrak{sl}_2$ (the oscillator / metaplectic triple)

$$E = \tfrac{1}{2}x^2, \qquad F = -\tfrac{1}{2}\frac{d^2}{dx^2}, \qquad H = x\frac{d}{dx} + \tfrac{1}{2},$$

with $[H, E] = 2E,\ [H, F] = -2F,\ [E, F] = H$ and, crucially, $\left[H, \tfrac{d}{dx}\right] = -\tfrac{d}{dx}$.

Key idea (The derivative is a lowering operator). In this $\mathfrak{sl}_2$ the derivative d/dx has *weight* -1 (it lowers), x has weight $+1$ (it raises), and $x\,d/dx$ is the *Cartan* (weight-counting) operator. The Stone–von Neumann theorem says this representation is essentially unique; that uniqueness is the precise reason the derivative is "canonical" — there is only one such operator up to equivalence.

10.3 Scaling is the Cartan grading; reduction is projection

The §2 scaling generator is now revealed as a Cartan operator that *grades* jet space by weight. For Burgers' scaling $X = 2t\,\partial_t + x\,\partial_x - u\,\partial_u$ assign

$$w(t) = 2, \qquad w(x) = 1, \qquad w(u) = -1,$$

so that each derivative subtracts the weight of the variable differentiated: $w(u_t) = w(u) - w(t) = -3$, $w(u\,u_x) = w(u) + w(u) - w(x) = -3$, $w(u_{xx}) = w(u) - 2w(x) = -3$. All three terms of $u_t + uu_x - \nu u_{xx}$ share weight -3: Burgers is **weight-homogeneous**, living in a single graded block.

Key idea (Reduction = isotypic projection). Aligning the jet basis with the scaling symmetry decomposes the differential-polynomial ring into *weight spaces* $R = \bigoplus_\lambda R_\lambda$ (the isotypic components of the prolonged action). A group-invariant (similarity) solution lives in the weight-0 piece. Hence the "variable compression" of §4 is literally the **projection π_0 onto the invariant subspace**, and the similarity variable z is a weight-0 coordinate.

10.4 Prolongation is a representation; invariants are highest-weight vectors

Prolongation respects the bracket,

$$\mathrm{pr}\,[X, Y] = [\,\mathrm{pr}\,X,\ \mathrm{pr}\,Y\,],$$

so $\mathrm{pr}^{(k)}$ is a Lie-algebra homomorphism $\mathfrak{g} \to \mathrm{Vect}(J^k)$ — a *representation*. The invariance condition (1), $\mathrm{pr}^{(k)}X(\Delta)|_{\Delta=0} = 0$, then says the solution set is an *invariant subspace*. A differential invariant ($\mathrm{pr}^{(k)}X(I) = 0$) is a highest-weight / Casimir-type vector, and the Lie–Tresse theorem says all invariants are generated by finitely many of them under an *invariant differentiation* operator — an equivariant derivative, the literal fusion of d/dx and the symmetry.

10.5 Fundamental PDEs realise classical algebras; ML equivariance is the same statement

Remark 10.1 (Heat, wave, and Green's functions). The heat equation $u_t = u_{xx}$ realises the Schrödinger algebra $\mathfrak{sl}_2 \ltimes$ Heisenberg (translations are momenta, dilation is the $\mathfrak{sl}_2$ Cartan, the

projective generator of Example 3.1 is the special-conformal element). The wave/Laplace equations realise the conformal algebra $\mathfrak{so}(p+1, q+1)$. A fundamental solution (heat kernel, Green's function) is a distinguished lowest-weight/resolvent vector. For *linear* PDEs this is the D-module picture: a PDE is a module $\mathcal{D}/\mathcal{D}\cdot P$ over the ring of differential operators, and the Riemann–Hilbert correspondence relates regular holonomic D-modules to monodromy representations of π_1 (derivatives $\leftrightarrow$ symmetry, once more).

Verdict. "Symmetry of a PDE" and "equivariance of a network" are the *same* equation, $\rho_{\text{out}}(g)\,\Phi = \Phi\,\rho_{\text{in}}(g)$ (§2.4). The findings of this section are not decoration: they explain *why* the CCS-SYM+ pipeline can reduce (project onto weight 0), close (the reduced operator is a clean $\mathfrak{sl}_2/D$-module object), and correct (an invariant-coordinate network is equivariant by construction, no loss term needed).

11 Exchange diagrams (commutative diagrams) for the basic definitions

A *commutative* (here "exchange") diagram says: *two routes give the same answer.* Each square below encodes one definition from §2–§10; reading both paths and checking they agree is a fast way to internalise the definition. (Diagrams use `amscd`; arrows are labelled above/below or beside.)

(D1) The shared atom: d/dx is the Lie derivative along ∂_x.

$$\begin{array}{ccc} C^\infty(M) & \xrightarrow{\mathcal{L}_X} & C^\infty(M) \\ \| & & \| \\ C^\infty(M) & \xrightarrow[D]{} & C^\infty(M) \end{array}$$

Both rows are derivations (each obeys Leibniz, $D(fg) = Df\,g + f\,Dg$); choosing $X = \partial_x$ makes the top row the ordinary derivative. One class of objects, not two.

(D2) Generator $\to$ flow (Lie's exponential).

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\exp} & G \\ \downarrow{\scriptstyle [\,\cdot\,,\cdot\,]} & & \downarrow{\scriptstyle \text{group law}} \\ X & \xrightarrow[t\mapsto \exp(tX)]{} & \{g_t\},\quad g_s g_t = g_{s+t} \end{array}$$

Addition in the algebra becomes composition in the group; this is the engine of §2.2.

(D3) Hopf duality: primitive $\leftrightarrow$ group-like.

$$\begin{array}{ccc} \text{Derivation } D & \xrightarrow{\exp} & \text{Automorphism } e^{tD} \\ \downarrow{\scriptstyle \Delta(D)=D\otimes 1+1\otimes D} & & \downarrow{\scriptstyle \Delta(e^{tD})=e^{tD}\otimes e^{tD}} \\ \text{primitive element} & \xleftarrow[\log]{} & \text{group-like element} \end{array}$$

(D4) Prolongation is a Lie-algebra homomorphism.

$$\begin{array}{ccc} \mathfrak{g}\times\mathfrak{g} & \xrightarrow{\mathrm{pr}^{(k)}\times\mathrm{pr}^{(k)}} & \mathrm{Vect}(J^k)\times\mathrm{Vect}(J^k) \\ \downarrow{\scriptstyle [\cdot,\cdot]_{\mathfrak{g}}} & & \downarrow{\scriptstyle [\cdot,\cdot]} \\ \mathfrak{g} & \xrightarrow[\mathrm{pr}^{(k)}]{} & \mathrm{Vect}(J^k) \end{array}$$

The square commutes: prolonging the bracket equals bracketing the prolongations.

(D5) Equivariance = intertwiner (PDE symmetry = ML equivariance).

$$\begin{array}{ccc} \mathbf{x} & \xrightarrow{\Phi} & \Phi(\mathbf{x}) \\ \downarrow{\scriptstyle \rho_{\mathrm{in}}(g)} & & \downarrow{\scriptstyle \rho_{\mathrm{out}}(g)} \\ \rho_{\mathrm{in}}(g)\,\mathbf{x} & \xrightarrow[\Phi]{} & \rho_{\mathrm{out}}(g)\,\Phi(\mathbf{x}) \end{array}$$

Φ is equivariant iff this square commutes: $\rho_{\mathrm{out}}(g)\,\Phi = \Phi\,\rho_{\mathrm{in}}(g)$.

(D6) Symmetries preserve the Cartan distribution.

$$\begin{array}{ccc} \text{generator } \mathrm{pr}^{(k)}X & \xrightarrow{[\,\cdot\,,D_i\,]} & [\mathrm{pr}^{(k)}X,\,D_i] \\ \downarrow{\scriptstyle \text{acts on } J^k} & & \downarrow{\scriptstyle \in \mathrm{span}(D_j)} \\ \mathcal{C}=\ker(\theta^0,\theta^1,\dots) & \xrightarrow[\text{preserved}]{} & \text{solution set invariant} \end{array}$$

The contact forms $\theta^0 = du - u_x\,dx - u_t\,dt,\ \dots$ cut out $\mathcal{C} = \mathrm{span}(D_x, D_t)$; symmetries keep $\mathcal{C}$ inside itself, so they send solutions to solutions.

(D7) The $\mathfrak{sl}_2$ weight ladder (derivative lowers, x raises).

$$\begin{array}{ccccc} V_{w+1} & \xrightarrow{d/dx} & V_w & \xrightarrow{d/dx} & V_{w-1} \\ \uparrow{\scriptstyle x} & & \uparrow{\scriptstyle x} & & \uparrow{\scriptstyle x} \\ V_w & = & V_w & = & V_w \end{array}$$

$x\,d/dx$ reads off the weight w; d/dx steps down, x steps up. Scaling symmetry sorts jet polynomials into these rungs, and invariants sit on the $w = 0$ rung.

(D8) Weight-homogeneity of Burgers and projection to invariants.

$$\begin{array}{ccc} \mathbb{R}[x,t,u,u_x,u_t,u_{xx},\dots] & \xrightarrow{\oplus_\lambda} & \bigoplus_\lambda R_\lambda \\ \downarrow{\scriptstyle u_t+uu_x-\nu u_{xx}} & & \downarrow{\scriptstyle \text{all terms weight } -3} \\ \text{solution set} & \xrightarrow[\pi_0]{} & R_0 \text{ (invariant / weight 0)} \end{array}$$

(D9) Riemann–Hilbert: D-modules → monodromy.

$$\begin{array}{ccc} \text{regular holonomic } D\text{-module } \mathcal{D}/\mathcal{D}P & \xrightarrow{\text{Sol}} & \text{local system} \\ \Big\downarrow{\scriptstyle \text{module over } \mathbb{R}[\partial_x,\partial_t,\dots]} & & \Big\downarrow \\ \text{linear PDE solutions} & \xrightarrow[\approx]{} & \text{monodromy rep of } \pi_1 \end{array}$$

(D10) The CCS-SYM+ pipeline as a single commuting chain.

$$\text{PDE } \Delta = 0 \xrightarrow{\text{reduce}} \text{ODE } F[U] = 0 \xrightarrow{\text{HAM}} U_M(z;\hbar) \xrightarrow{\text{certify}} B_M(\varepsilon) \xrightarrow{\text{correct}} u + \epsilon_{\text{PINN}}$$

Symmetry compresses; the homotopy-analysis series closes; the ratio/a-posteriori bound certifies; the small equivariant PINN corrects the remainder.

Remark 11.1 (Why `amscd` and not `tikz-cd`). These diagrams use only horizontal/vertical arrows, which `amscd` (bundled with `amsmath`) draws with no extra package. If you later want diagonal or curved labelled arrows, add `\usepackage{tikz-cd}` and translate each `CD` block into a `tikzcd` environment; the mathematics is unchanged.

12 Foundations: sheaves, schemes, jets, D-modules

This section supplies, gently, the "language layer (B)" that §7 flagged as optional enrichment. None of it is needed to *run* a reduction, but it makes precise *why* the constructions of §10 are the right ones. Read for vocabulary, not for proofs.

12.1 Presheaves and sheaves: local-to-global

Definition 12.1 (Presheaf and sheaf)**.** A **presheaf** $\mathcal{F}$ on a space X assigns to each open $U \subseteq X$ a set (or group, ring) $\mathcal{F}(U)$ of "sections", with restriction maps $\rho_{V\subseteq U} : \mathcal{F}(U) \to \mathcal{F}(V)$ that compose correctly and fix $\mathcal{F}(U)$ for $V = U$. It is a **sheaf** if sections that agree on overlaps glue to a *unique* global section over any open cover.

Example 12.1 (Sheaves you already use)**.** Smooth functions $U \mapsto C^\infty(U)$; differential k-forms $U \mapsto \Omega^k(U)$; and the *solution sheaf* of a PDE, $U \mapsto \{u : U \to \mathbb{R} \text{ with } \Delta = 0 \text{ on } U\}$. The last is why "the space of solutions" is local data, not a single global vector space (boundary/initial conditions vary from patch to patch).

Definition 12.2 (Stalk, ringed and locally ringed space)**.** The **stalk** $\mathcal{F}_x = \varinjlim_{U\ni x} \mathcal{F}(U)$ collects *germs* at x. A **ringed space** $(X, \mathcal{O})$ carries a sheaf of rings $\mathcal{O}$ (the *structure sheaf*); it is **locally ringed** if every stalk $\mathcal{O}_x$ has a unique maximal ideal $\mathfrak{m}_x$.

Remark 12.1. A smooth manifold M with $\mathcal{O} = C^\infty_M$ is a locally ringed space: $\mathfrak{m}_x$ is the germs vanishing at x. This is the bridge from analysis to the algebraic picture below.

12.2 Affine schemes: "geometry is commutative algebra"

Definition 12.3 (Affine scheme)**.** For a commutative ring R, the **spectrum** $\operatorname{Spec} R$ is the set of prime ideals with the Zariski topology (closed sets $V(I) = \{\mathfrak{p} : I \subseteq \mathfrak{p}\}$) and structure sheaf determined by $\mathcal{O}(D(f)) = R[1/f]$ on basic opens $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$. The pair $(\operatorname{Spec} R, \mathcal{O})$ is a locally ringed space; morphisms of schemes are (contravariantly) ring homomorphisms.

Example 12.2 (The jet ring is a scheme; a PDE is a closed subscheme)**.** The differential-polynomial ring $R = \mathbb{R}[x, t, u, u_x, u_t, u_{xx}, u_{xt}, \dots]$ encodes "all possible jets" of $u(x,t)$. A PDE $\Delta = 0$ cuts out the closed subscheme $V(\Delta) \subseteq \operatorname{Spec} R$, and a symmetry is an automorphism of $\operatorname{Spec} R$ preserving $V(\Delta)$. This is the algebraic twin of the geometric "invariant subvariety" of §10.

12.3 Jet bundles and jet schemes: where derivatives live

Definition 12.4 (Jet bundle and Cartan distribution)**.** For a field $u : \mathbb{R}^2 \to \mathbb{R}$ the k-jet bundle J^k has coordinates $(x, t, u, u_x, u_t, u_{xx}, u_{xt}, u_{tt}, \dots)$. The **Cartan (contact) distribution** $\mathcal{C} \subseteq TJ^k$ is spanned by the *total derivatives*

$$D_x = \partial_x + u_x\partial_u + u_{xx}\partial_{u_x} + u_{xt}\partial_{u_t} + \cdots, \qquad D_t = \partial_t + u_t\partial_u + u_{xt}\partial_{u_x} + u_{tt}\partial_{u_t} + \cdots,$$

with $[D_x, D_t] = 0$ (mixed partials commute). It is cut out by the *contact forms* $\theta^0 = du - u_x\,dx - u_t\,dt$, $\theta^1 = du_x - u_{xx}\,dx - u_{xt}\,dt, \dots$.

Key idea (A PDE is a pair (subvariety, distribution))*.* A solution $u(x,t)$ lifts to a curve in J^k that (i) lies in $V(\Delta) = \{\Delta = 0\}$ and (ii) is an integral manifold of $\mathcal{C}$. Conversely an integral manifold of $\mathcal{C}$ inside $V(\Delta)$ projects to a solution. So "a PDE" is honestly the pair $(V(\Delta), \mathcal{C})$ — this is the geometric meaning of (1). (Algebraically, J^k is itself a scheme, the *jet scheme* Spec of the differential-polynomial ring, unifying the two viewpoints.)

12.4 Differential operators and D-modules

Definition 12.5 (Differential operator; D-module)**.** A **differential operator** of order $\le k$ is $D[f] = \sum_{|\alpha| \le k} a_\alpha(x)\partial^\alpha f$; these form a non-commutative ring $\mathcal{D}$ under composition. A **$\mathcal{D}$-module** is a module over $\mathcal{D}$; for a linear PDE $L[u] = 0$ the solution space $\operatorname{Sol}(L) = \{u : L[u] = 0\}$ is one.

Key idea (Solutions are representations)*.* If X is a symmetry generator then $\operatorname{pr}^{(k)} X$ commutes with L, so $\operatorname{Sol}(L)$ is a representation of $\mathfrak{g}$ (equivalently a $\mathcal{D}$-module). Borel–Weil–Bott is the statement that irreducible representations are realised exactly as such solution spaces. This is the rigorous form of "the heat kernel is a distinguished vector" from §10, and it underwrites the convergence *certificate*: the reduced linear operator is invertible in the right function space precisely because its D-module is well-behaved.

12.5 Hopf algebras and the primitive/group-like dictionary

Definition 12.6 (Hopf algebra; primitive and group-like)**.** A **Hopf algebra** is an algebra with a coproduct Δ, counit ε, and antipode S. In the enveloping algebra $U(\mathfrak{g})$ one sets $\Delta(X) = X \otimes 1 + 1 \otimes X$ for $X \in \mathfrak{g}$. An element is **primitive** if $\Delta(x) = x \otimes 1 + 1 \otimes x$, and **group-like** if $\Delta(g) = g \otimes g$.

This is the engine behind §10: every $X \in \mathfrak{g}$ is primitive (a derivation), every group element is group-like (a flow), and exp ferries one to the other (diagram D3). The relation $[\,d/dx,\, x\,] = 1$ and its $\mathfrak{sl}_2$ closure (diagram D7) live here too; Stone–von Neumann's uniqueness is what makes the derivative *the* canonical primitive.

12.6 Differential invariants, moving frames, and diffieties

Definition 12.7 (Differential invariant; invariant differentiation)**.** $I(u, u_x, u_t, \dots)$ is a **differential invariant** if $\operatorname{pr}^{(k)} g \cdot I = I$ for all g. An **invariant differentiation** $\bar{D}$ maps invariants to invariants; the Lie–Tresse theorem (Fels–Olver) says the whole algebra of invariants is finitely generated by base invariants together with $\bar{D}$ and its iterates.

Example 12.3 (Why a similarity reduction is automatic)**.** In invariant coordinates $(z, U(z))$ the action is trivial and $\bar{D} = d/dz$; the reduced ODE is then invariant *by construction*. An ML feature built from invariants (or ratios of them) is exactly equivariant with no auxiliary symmetry loss — the rigorous form of the §7 claim.

Definition 12.8 (Diffiety)**.** A **diffiety** is a PDE seen intrinsically: a quotient of a free differential algebra by a **differential ideal** (closed under all total derivatives). Vinogradov's *secondary calculus* does differential geometry on the diffiety itself; Noether's theorem and conservation laws appear as classes in its *variational bicomplex*.

Verdict. Sheaves, schemes, jets, D-modules, Hopf algebras and diffieties are not window dressing: they are the exact language in which the findings of §10 are theorems — the derivative is a primitive, scaling is a grading, reduction is projection onto weight 0, symmetries are automorphisms of the Cartan distribution, and a convergence certificate is a closedness statement about the solution algebra. For this thesis, *reading* this section is enough; *mastering* it is a later, optional investment.

Glossary

Lie group
: a group that is also a smooth manifold, with smooth multiplication and inversion; the continuous symmetries of a PDE (§2.3).

Homeomorphism
: a continuous bijection with continuous inverse; "same shape up to continuous deformation," no cutting or gluing.

Locally Euclidean
: every point has a neighbourhood homeomorphic to an open set of $\mathbb{R}^n$ — "up close it looks flat like $\mathbb{R}^n$"; the n is the dimension.

Chart / atlas
: a chart is a local coordinate map $\varphi : U \to \mathbb{R}^n$; an atlas is a set of charts covering M, glued by smooth transition maps.

(Smooth) manifold
: a Hausdorff, second-countable, locally Euclidean space whose chart transition maps are all C^∞; the space on which calculus makes sense.

Tangent space T_pM
: the vector space of velocities of curves through p; the Lie algebra is the tangent space at the identity, $\mathfrak{g} = T_eG$.

Closed-subgroup theorem
: (Cartan) every topologically closed subgroup of $GL(n, \mathbb{C})$ is automatically a smooth submanifold, hence a Lie group — why "closed" suffices in the matrix definition.

Lie algebra $\mathfrak{g}$
: the (finite-dimensional) space of infinitesimal generators with bracket $[X, Y] = XY - YX$; the linear shadow of the group.

Matrix (linear) Lie group
: a closed subgroup of $GL(n, \mathbb{C})$ (e.g. $SO(n)$, $SU(n)$); the concrete model in which exp, the bracket, and representations are plain linear algebra (Hall).

Matrix exponential
: $\exp(A) = \sum_k A^k/k!$; sends an algebra element (matrix) to a group element.

Representation
: a homomorphism $\rho : G \to GL(V)$; how the group acts linearly on a space V — the shared language of PDE symmetry and ML equivariance.

Generator
: a vector field $X = \xi\partial_x + \tau\partial_t + \phi\partial_u$ encoding "move a little".

Prolongation
: the extension of X to derivatives, needed to test PDEs.

Determining equations
: the linear, overdetermined system whose solution is $\mathfrak{g}$.

Invariant / similarity variable
: a function killed by X; the new single variable after reduction.

Reduction
: using a symmetry to lower the number of independent variables (PDE $\to$ ODE).

Optimal system
: one representative reduction per conjugacy class of subalgebra; the combinatorial catalogue of inequivalent reductions.

Equivariance
: an ML model commuting with a group action; "respecting the symmetry".

Finite group of Lie type
: a finite group over $\mathbb{F}_q$ (Chevalley group); *not* used here.

Derivation
: a linear map obeying Leibniz $D(fg) = Df\,g + f\,Dg$; both d/dx and a Lie generator $\mathcal{L}_X$ are derivations (§10).

Primitive / group-like
: the Hopf shadow of generator/flow: $\Delta(D) = D \otimes 1 + 1 \otimes D$ vs $\Delta(g) = g \otimes g$, linked by $\exp$ / $\log$.

Weight / grading (Cartan)
: the eigenvalue under the scaling operator $x\,d/dx$; jet monomials sort into weight spaces, invariants sit at weight 0.

Differential invariant
: a function with $\mathrm{pr}^{(k)}X(I) = 0$; finitely generated under invariant differentiation (Lie–Tresse).

Jet bundle / jet scheme
: the space J^k of coordinates $(x, t, u, u_x, \dots)$ where derivatives live; algebraically Spec of the differential-polynomial ring.

Cartan (contact) distribution
: the span of the total derivatives D_x, D_t on J^k; solutions are its integral manifolds inside $\{\Delta = 0\}$.

Sheaf
: data on open sets with restrictions that glue uniquely on covers; e.g. the solution sheaf of a PDE.

Scheme
: a locally ringed space locally of the form $(\operatorname{Spec} R, \mathcal{O})$; "geometry = commutative algebra".

D-module
: a module over the ring $\mathcal{D}$ of differential operators; a linear PDE is one, its solutions a representation of $\mathfrak{g}$.

Diffiety
: a PDE as a differential algebra modulo a differential ideal; home of Vinogradov's secondary calculus.

Riemann–Hilbert
: the correspondence from regular holonomic D-modules to monodromy representations of π_1.

Schrödinger / conformal algebra
: the maximal symmetry algebras of the heat ($\mathfrak{sl}_2 \ltimes$ Heisenberg) and wave/Laplace ($\mathfrak{so}(p{+}1, q{+}1)$) equations.

Companion file: the full research programme (thesis, methodology, phased plan, benchmarks, verified feasibility) is in `plan.txt`; the runnable computations referenced above are in `symm/`.

References for the advanced sections

References

[1] P. J. Olver, *Applications of Lie Groups to Differential Equations*, 2nd ed., Springer, 1993.

[2] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Springer, 2015.

[3] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., GTM 218, Springer, 2012 (manifolds, charts, tangent spaces; Lie groups in Ch. 7, 8, 20).

[4] A. M. Vinogradov, *Cohomological Analysis of PDEs and Secondary Calculus*, AMS, 2001; survey arXiv:2301.02038.

[5] Lie–Tresse finite generation of differential invariants; arXiv:2011.05101.

[6] C. Duval, M. Henkel, P. Horvathy, S. Rouhani, P. Zhang, *Schrödinger Symmetry: A Historical Review*, arXiv:2403.20316.

[7] J. Brandstetter, M. Welling, D. Worrall, *Lie Point Symmetry Data Augmentation for Neural PDE Solvers*, arXiv:2202.07643.

[8] J. Helwig et al., *Group Equivariant Fourier Neural Operators*, arXiv:2306.05697.

[9] J. Yang et al., *Discovering Symbolic Differential Equations with Symmetry Invariants*, arXiv:2505.12083.

[10] S. Ishii, *Jet schemes, arc spaces and the Nash problem*, arXiv:0704.3327.

[11] M. E. Sweedler, *Hopf Algebras*, Benjamin, 1969.

[12] M. H. Stone, On one-parameter unitary groups in Hilbert space, Ann. Math. 33 (1932) 643–648.